UNDULI SENADHEERA

Mechanical Engineering Undergraduate

Specialization in the Aeronautical Stream

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PROFILE

Third-year Mechanical Engineering undergraduate specializing in the Aeronautical Stream at the University of Moratuwa, with hands-on project experience spanning aerodynamic design and analysis, CAD (SolidWorks, XFOIL/XFLR5, ANSYS), composite and precision manufacturing, and reverse engineering / materials identification. Comfortable working across electronics and automation (Python, C++, Arduino/ESP32).

Strong communicator with leadership experience across multiple engineering student chapters.

EDUCATION

BSc (Hons) in Mechanical Engineering – Aeronautical Stream, University of Moratuwa 2024 – Present

- CGPA (up to Semester 4): 3.91

St. Joseph's Balika Vidyalaya, Kegalle — G.C.E. Advanced Level 2022

- Z-score: 2.1481 — Combined Mathematics (A), Chemistry (A), Physics (B)
- Island rank: 694

St. Joseph's Balika Vidyalaya, Kegalle — G.C.E. Ordinary Level 2019

- 9 A's (Distinction passes)

SKILLS

Technical & Software

- Aerodynamic analysis (XFOIL, XFLR5)
- CAD & 3D modelling (SolidWorks)
- Basic FEA (ANSYS)
- Composite manufacturing (carbon fiber)
- Machining, welding & workshop practice
- Electronics & automation (Arduino, ESP32)
- Programming: Python, C++, MATLAB
- Microsoft Office

Professional

- Leadership
- Public speaking & compering
- Programme management
- Teamwork & collaboration
- Critical thinking & problem solving
- English & Sinhala (fluent)

PROJECTS

Glider Aerodynamic Design — Group Project Sep 2025 – Dec 2025

Designed, analyzed, and manufactured a 60 g small-scale glider optimized for a cruise speed of 10 m/s, focusing on aerodynamic performance and structural stability.

- Performed aerodynamic analysis and airfoil selection for the wing using MATLAB and XFOIL.
- Determined wing and tail placement for static stability using XFLR5.
- Designed the fuselage in SolidWorks.

- Contributed to manufacturing the airframe, laminating the wing and tail skins with heat-shrink covering film using a heated iron.

Falcon E Racing — Formula Student Dec 2025 – Present

Selected team member, Formula Student Spain 2026, contributing to the aerodynamics subsystem.

- Designed the rear-wing mounting bracket in SolidWorks to satisfy rear-wing compliance requirements in the Formula Student Rulebook 2026.
- Manufactured the front wing, rear wing, side/rear diffusers, and all body panels of the vehicle in carbon fiber using wet-layup vacuum bagging.
- Designing the Rear wing for the next car E3.

Smart Cooking System — Group Project Jan 2025 – May 2025

Designed and developed a semi-automated smart cooking system integrated with a mobile app, with cross-device communication via Bluetooth.

- Built an Arduino- and .NET MAUI-based system that measures the mass of a curry's main ingredient via load cell, then calculates and dispenses proportional quantities of spices, other ingredients, water, and coconut milk relative to 1 kg of main ingredient (demo used water as the reference liquid). Tools: Visual Studio (C++), .NET MAUI, SQLite.
- Calibrated the load cell and coded the mass-measurement and calculation logic; integrated the full system using ESP32 and Arduino. Tools: Arduino IDE, HX711 load sensor, ultrasonic sensors, stepper motors.
- Set up cross-device communication between the system and the mobile application via Bluetooth (ESP32, HC-05).

Kick Scooter Manufacturing — Group Project Jan 2025 – May 2025

Designed and fabricated a kick scooter from scratch in the university workshops.

- Selected materials for each component based on functionality, cost, and manufacturability.
- Gained hands-on experience in machining, welding, and woodworking as part of the manufacturing process.

Reverse Engineering — Hand Mixer (Group Project) Jan 2026 – May 2026

Reverse-engineered a hand mixer, identifying the materials used for each component and the manufacturing process, and proposed improvements through laboratory testing and CAD modelling.

- Analyzed the housing of the hand mixer; the material was identified as ABS plastic through material testing.
- Modelled the hand mixer housing design using SolidWorks.
- Identified possible manufacturing methods for the housing.
- Proposed design improvements to enhance the ergonomics of the device.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **Affiliated Student Member** — Institution of Mechanical Engineers (IMechE)
- **Team Member (Aerodynamics Subsystem)** Falcon E Racing, University of Moratuwa (2025 - Present)
- **Editorial Pillar Head** — IMechE Student Chapter, University of Moratuwa (2025/26)
- **Volunteering Member** — IEEE Student Branch of University of Moratuwa (2024 – 2026)
  - **Financial Committee Member** — Mora Foresight 2.0 & 3.0 (2024/25)
  - **Selection Committee Lead** — Mora Foresight 4.0 (2026)
  - **Editorial Committee Member** — MERCon 2025 (IEEE Student Branch, University of Moratuwa)
- **Member of Mora Wings (Compering)** — IESL Student Chapter, University of Moratuwa (2025/26)
- **Editorial Pillar Member** — Robotics & Automation Student Chapter, IEEE Student Branch, University of Moratuwa (2024/25)

- **Financial Committee Member** — Leo Club, University of Moratuwa (2024/25)
- **Rotaractor** — Rotaract Club of University of Moratuwa (Inducted 2025)

ACHIEVEMENTS

- **Dean's List** — Semester 1, 2, 3, 4
- **Speak Out for Engineering (SOFE) 2025** — Sri Lankan Finalist, IMechE Sri Lanka
  - Topic: "Nature is the Blueprint of Engineering" (Biomimicry)
  - Researched and presented how biomimicry translates natural systems into engineering solutions, using case studies including Velcro (inspired by burdock burrs), humpback-whale-flipper-inspired wind turbine blades, mangrove/coral-root-inspired soil anchoring systems, and termite-mound-inspired smart ventilation systems.
- **Speech Olympiad XVII** — Reserve Finalist, Gavel Club of University of Moratuwa (2024)

REFERENCES

Dr. L. U. Subasinghe

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I hereby certify that the above particulars given by me are true and accurate to the best of my knowledge.

17/08/2026 ………...……………………………….

Date Signature